RESERVA HUARAVITO

HUARAVITO LTEMS

Evening Microclimatic Stabilization Event

Date: February 26, 2026
Location: (10.104583, -84.564639)
Piedades Sur, San Ramón, Costa Rica
Reserva Huaravito Long-Term Ecological Monitoring Site (LTEMS)

Technical Note

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1. Background

The Bosque Sagrado Toyopán (≈ 4,500 m²) functions as the structural core of the Huaravito LTEMS. The site is characterized by:

- Closed canopy dominated by Hauya elegans (estimated basal area 21–34 m²/ha)
- Deep organic soil layer (~20 cm mean depth)
- High epiphytic load (>70% trunk surface coverage)
- Surrounding anthropogenic matrix (pastures, abandoned clay extraction site, secondary growth)

The objective of this note is to document a same-day thermal stabilization event following a high-radiation morning, establishing a baseline for comparison with anticipated Saharan dust intrusion events (early March 2026).

2. External Regional Conditions (26 February 2026)

Regional meteorological conditions for Piedades Sur, based on available data:

Parameter	Value
Mean daytime temperature	29.7°C
Evening temperature (~20:00)	~22°C
Relative humidity (daytime)	82%
Wind conditions	Low to moderate
Sky condition	Cloud cover increasing toward evening

Table 1. Regional meteorological conditions representing open or semi-open terrain outside forest core.

These values represent ambient conditions in the surrounding matrix (agricultural lands, open areas) and serve as the comparative baseline for evaluating forest interior microclimate.

3. Internal Microclimatic Observations (Toyopán Core)

3.1 Sensor Location and Protocol

Temperature monitoring was conducted using a mobile digital sensor positioned at 1.5 m height within the forest understory, adjacent to a residential structure but beneath closed canopy (southeast sector, internal road to My House Hill). The sensor is exposed to forest air mass, not indoor conditions. This location corresponds to the edge-to-core transition zone, approximately 25 m interior from the southeastern boundary.

3.2 Temperature Measurements

Time	Location	Temp (°C)	Condition
Morning	External	29.7	High radiation
17:00	Forest core	19.6	Cloud cover increasing
20:00	Forest core	**17.0**	Stabilization
Evening	External	~22	Slow cooling

Table 2. Temperature measurements showing forest vs. external conditions. Highlighted cell indicates minimum recorded temperature.

Recorded temperature range (forest core): 17.0–18.3°C

3.3 Temperature Differential (ΔT) Calculation

Comparing evening forest core temperature with external ambient:

External (~20:00): 22°C

Forest core (20:00): 17.0°C

ΔT ≈ -5.0°C (forest cooler)

Note: This represents a conservative estimate based on regional reference temperature. The actual external temperature at the forest boundary (abandoned quarry, edge zones) may have been higher, suggesting a potentially larger ΔT.

3.4 Cooling Rate Analysis

Forest interior exhibited rapid cooling during the diurnal-to-nocturnal transition:

Forest cooling: -2.6°C over 3 hours (17:00 → 20:00)

Cooling rate: -0.87°C/hour

In contrast, external temperatures declined more gradually:

External cooling: -7.7°C over ~8 hours (morning 29.7°C → evening 22°C)

Cooling rate: -0.96°C/hour

While cooling rates are comparable, the forest stabilized at a substantially lower absolute temperature (17.0°C vs. 22°C), indicating effective thermal buffering.

4. Event Description

Morning conditions were characterized by high solar radiation and warm ambient temperatures (29.7°C). By late afternoon and into evening:

- Cloud cover increased, reducing direct solar radiation
- Wind velocity decreased, minimizing convective mixing
- Forest interior temperature dropped steadily from 19.6°C to 17.0°C
- External temperature decreased more slowly (morning 29.7°C → evening 22°C)

The forest core exhibited rapid stabilization toward a 17°C baseline, consistent with nocturnal cooling dynamics in thermally buffered forest systems.

5. Observed Patterns

5.1 Thermal Decoupling

Despite daytime heating in the regional matrix (29.7°C), the forest interior maintained a consistent sub-ambient temperature throughout the observation period. Evening stabilization occurred without external forcing (no rainfall recorded), suggesting endogenous cooling mechanisms driven by forest structure and evapotranspiration.

5.2 Reduced Thermal Variability

Preliminary qualitative observation: External conditions exhibited broader thermal fluctuation (morning 29.7°C → evening 22°C; range 7.7°C), while internal readings remained within a narrow range (17.0–19.6°C; range 2.6°C). This reduced variability suggests active thermal buffering capacity.

5.3 Comparison with Historical Baseline

The observed evening stabilization at 17.0–18.3°C is slightly cooler than the established four-year baseline (2022–2025) of 19.5°C ± 0.2°C. This may reflect:

- Nocturnal cooling dynamics (measurements taken during evening hours rather than mid-day)
- Spatial variation (southeast sector edge-transition zone vs. western core “The Shrine”)
- Seasonal variation (late February = end of dry season with potential increased evaporative cooling)

Despite the lower absolute values, the observed sub-ambient stabilization is consistent with documented biogenic cooling patterns.

6. Proposed Mechanisms (Preliminary)

Based on known structural characteristics of the Toyopán forest, the observed thermal stabilization is hypothesized to result from:

- Dense canopy reducing shortwave radiation input: Closed Hauya elegans canopy (basal area 21–34 m²/ha) intercepts solar radiation, minimizing heat gain at ground level.
- High evapotranspiration from mature trees: Large-diameter H. elegans with extensive crown volumes contribute substantial latent heat flux through transpiration.
- Organic soil layer thermal inertia: Deep organic layer (~20 cm) retains moisture and buffers against rapid temperature fluctuations.
- Reduced convective mixing under closed canopy: Stratified air masses beneath dense canopy minimize advective heat transport from warmer external zones.

These mechanisms are consistent with the previously described biogenic cooling hypothesis and the proposed orographic fog interception model (northeast boundary cold air influx).

7. Interpretation

> **The evening data support the hypothesis that the Toyopán forest core operates as a microclimatic stabilization node capable of maintaining sub-ambient conditions (-5.0°C relative to regional ambient) despite high daytime radiation and warm matrix temperatures.**

However, this note represents a single-event documentation and does not yet constitute longitudinal statistical validation. The observation should be interpreted as: (a) confirmation that thermal decoupling persists under variable weather conditions; (b) establishment of a baseline for comparison with anticipated Saharan dust intrusion events; and (c) motivation for continued systematic monitoring.

8. Limitations

- Single sensor location: No simultaneous spatial replication (edge vs. core)
- No synchronized external sensor: Regional reference temperature (~22°C) is estimated from meteorological data, not direct on-site measurement
- Short temporal window: Observations cover 3-hour period (17:00–20:00) rather than full diurnal cycle
- Edge-transition zone measurement: Sensor positioned in southeast sector transition zone rather than deepest forest core (“The Shrine”), which may exhibit even greater cooling
- Visual cloud cover assessment: No quantitative measurement of sky conditions or solar radiation flux

9. Next Steps

9.1 Immediate Actions

- Replicate measurements for 2–3 additional days (pre-Saharan dust baseline establishment)
- Document all weather conditions (cloud cover, wind, precipitation) qualitatively
- Prepare for Saharan dust monitoring protocol (anticipated early March 2026)

9.2 Medium-Term Monitoring Protocol

- Install synchronized external sensor in open area (abandoned quarry) for precise ΔT computation
- Collect 90-day continuous dataset spanning dry-to-wet season transition
- Calculate mean ΔT, diurnal amplitude, and interday variance
- Perform comparative statistical analysis (edge vs. core; forest vs. matrix)

9.3 Event-Based Monitoring

- Document Saharan dust intrusion event (temperature, humidity, visibility, sky color)
- Compare ΔT during dust event vs. baseline (hypothesis: ΔT increases under external stress)
- Assess forest resilience under regional climatic perturbation

10. Provisional Conclusion

The 26 February 2026 stabilization event provides observational evidence consistent with:

- Thermal buffering capacity: Forest maintained 17.0–19.6°C while external temperatures reached 29.7°C (daytime) and 22°C (evening)
- Sub-ambient stabilization: ΔT ≈ -5.0°C evening differential demonstrates sustained cooling effect
- Rapid post-radiation recovery: Forest cooled -2.6°C in 3 hours following high-radiation morning

This observation supports continuation of the Sentinel Microclimate Protocol under the Huaravito LTEMS framework. Future work will focus on: (a) establishing multi-day baseline prior to Saharan dust event; (b) deploying synchronized sensor network; and (c) quantifying seasonal and inter-annual thermal variance.

Acknowledgment

This observation contributes to the ongoing microclimatic monitoring program under the Sentinel Protocol. Temperature data will be integrated into the LTEMS Raw Datasets collection and compared with the established four-year baseline (2022–2025). The anticipated Saharan dust intrusion event (early March 2026) will provide a natural experiment to assess forest microclimate resilience under regional climatic stress.

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